

Supporting Information

Myke Vital Sao Temgoua Jean-Pierre Tchapel Njafa Serge Guy Nana Engo
Penabei Samafou Wilfred Fon Mbacham

1 Orthogonal LISH mechanism-of-action benchmark

This section provides the complete protocol and results of the phenotype-only LISH benchmark that is summarized in the main text. It is an orthogonal pharmacological reference, not a validation of the molecular representations studied in the main benchmark.

1.1 Data and preparation

We used the public LISH MoA training release as a separate, phenotype-only benchmark; the public challenge repository is available at https://github.com/LISHarvard/moa_challenge (accessed 12 August 2026). The preparation retained 772 gene-expression features, 100 cell-viability features, and 3 experimental-condition summary columns (exposure time, high-dose fraction, treatment-type fraction), and aggregated repeated assay observations to the drug level.

The scored endpoint comprised 206 MoA labels for 3 289 drugs; labels were defined at the drug level from observed MoA-associated annotations. Because the public release used here did not contain a versioned drug-SMILES mapping, the benchmark was not used to train or evaluate molecular fingerprints, GNNs, or transformers. A prespecified sensitivity analysis excluded vehicle-control samples, yielding 3 288 drugs. All fold records were audited for missing or non-finite metrics before interpretation. The endpoint is described as MoA-associated prediction because it does not establish causal target engagement [1].

1.2 Protocol

We fitted an unweighted per-label logistic model independently for each of the 206 labels, evaluated with five random seeds and five drug-grouped folds, and reported mean column-wise log loss as the primary metric with macro-AUPRC, macro-AUROC, mean Brier score, and expected calibration error as secondary metrics.

1.3 Results

With vehicle controls retained, the phenotype-only baseline achieved the metrics reported in Table 1 across 25 fold-seed evaluations. Removing vehicle controls changed macro-AUROC by only -0.0018 , and the other metrics moved by less than 0.002 in absolute value, indicating that the phenotype-driven baseline is robust to control exclusion.

1.4 Interpretation boundary

These metrics summarize the phenotype-only reference; they are **not numerically comparable** with the molecular ROC-AUC results of the main manuscript because the task, labels, features, aggregation unit, and primary metric differ. The labels indicate MoA-associated prediction rather than causal target engagement. The LISH analysis therefore serves as an orthogonal pharmacological reference and control-sensitivity analysis, not as a molecular-model validation or evidence of causal mechanism.

Table 1: Phenotype-only LISH baseline metrics, mean across 25 fold-seed evaluations (five seeds $\times$ five drug-grouped folds).

Metric	With controls	Without controls	Δ
Mean column-wise log loss	0.023 78	0.023 89	+0.000 11
Macro-AUROC	0.643 5	0.641 7	−0.001 77
Macro-AUPRC	0.142 8	0.141 2	−0.001 52
Mean Brier score	0.003 74	0.003 75	+0.000 01
Expected calibration error	0.003 76	0.003 81	+0.000 05

2 Post-hoc calibration metrics for the molecular benchmark

Calibration metrics were computed post-hoc from the archived fold-level test predictions of the extended campaign (five seeds $\times$ five folds per configuration; 99 180 pooled test records per configuration). No model was retrained; the estimates therefore describe the calibration of the deployed checkpoints, pooled over seeds and folds, and are reported as a pooled-calibration estimand. Expected calibration error (ECE) and maximum calibration error (MCE) use 10 equal-width bins; the Brier score is the mean squared error between predicted probability and label.

Table 2: Post-hoc calibration metrics (pooled over 25 fold-seed evaluations per configuration). ECE and MCE rise from the random split to the scaffold and novel-partition splits for every model family, indicating that confidence degrades under scaffold shift even when ranking (ROC-AUC) is only moderately affected.

Split	Model	ECE	MCE	Brier
Canonical random	GIN	0.029 0	0.069 9	0.102 55
	GIN-TFP (native)	0.034 4	0.071 6	0.105 24
	GIN-TNE (native)	0.036 7	0.093 6	0.112 86
	ChemBERTa	0.057 8	0.144 0	0.104 66
Canonical scaffold	GIN	0.065 4	0.162 3	0.148 59
	GIN-TFP (native)	0.074 1	0.175 4	0.147 69
	GIN-TNE (native)	0.080 7	0.194 6	0.151 08
	ChemBERTa	0.122 5	0.254 6	0.169 09
Novel partitions (range)	GIN	0.0777–0.0888	0.1814–0.1854	0.15359–0.15919
	GIN-TFP (native)	0.0741–0.0895	0.1737–0.1890	0.14922–0.15564
	GIN-TNE (native)	0.0798–0.0820	0.1813–0.2078	0.15305–0.15539
	ChemBERTa	0.1172–0.1320	0.2401–0.2715	0.16699–0.17508

The degradation is consistent across families: ECE roughly doubles from the random split to the scaffold split and widens further on the novel partitions, and ChemBERTa — the arm with the largest ROC-AUC fold-level spread — also shows the largest calibration error under scaffold shift. These estimates are secondary robustness evidence; they do not alter the canonical ROC-AUC results and are not presented as calibrated probabilities for prospective use. Reliability-diagram JSON per configuration and the machine-readable summary are archived in [results/calibration_20260827/](https://results.calibration_20260827/).

References

- [1] Maria-Anna Trapotsi, Layla Hosseini-Gerami, and Andreas Bender. Computational analyses of mechanism of action (moa): data, methods and integration. *RSC Chemical Biology*, 3(2):

170–200, 2022. doi: 10.1039/D1CB00069A.